



Original Research Article

Understanding recruitment limitations in a critically endangered species: The story of the iconic Cape cedar (*Widdringtonia wallichii*)



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ABSTRACT

Demographic modelling predicts the critically endangered Clanwilliam cedar (*Widdringtonia wallichii*), an iconic conifer of South Africa, to be headed for extinction. Due to an apparent mismatch in life-history traits with the local fire regime, such as time to reproductive maturity exceeding mean fire return intervals and no clear seed dispersal agent, it is considered poorly adapted to its fire-prone environment.

Here we focus on two key demographic knowledge gaps, seed dispersal and germination, and suggest cedars are highly adapted to wildfires. Our natural history observations show that seeds are scatterhoarded by the Cape spiny mouse (*Acomys subspinosus*) to safe sites that are optimal for germination. In many cedar sites spiny mice are absent, scatterhoarding has collapsed, and seed predation by the Namaqua rock rat (*Micaelamys namaquensis*) is most frequently observed. Additionally, seeds are fire-stimulated. However, 85% of trees experienced ca. 70% of their fires before annual seed fall, limiting recruitment opportunities.

Our results suggest the cedar is strongly fire-adapted, and that the main demographic hurdle presently is not high adult mortality in fires but poor post-fire recruitment. Consequently, conservation emphasis should be on increasing recruitment rather than mitigating adult mortality. We suggest the following conservation measures: i) prescribed fires after seed fall (i.e. late autumn) to allow for scatterhoarding by rodents, ii) artificial burial of pre-heated seeds in burned stands in late autumn to optimise recruitment, and iii) planting locally dense stands of seeds/seedlings to guarantee predator satiation of seeds and thus strongly establish the scatterhoarding mutualism required for effective recruitment.

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1. Introduction

Large, long-lived trees are keystone structures in several ecosystems as they play critical ecological roles, including providing abundant food, such as seeds, fruits, nectar and flowers, to numerous animals (Lindenmayer et al., 2012). They are threatened globally and are disproportionately vulnerable to drivers of mortality (Bennett et al., 2015). Mortality and, importantly, recruitment are key components in any life-cycle and need to be considered together to understand population dynamics. Understanding a plant's full reproductive ecology can provide insights that allow for more effective conservation of populations (Wang and Smith, 2002), as efforts to address threats can be aligned with key life-history strategies. These key demographic knowledge gaps are particularly important to recognize for long-lived, sporadically recruiting plants, such as Baobabs (*Adansonia digitata*) and Joshua trees (*Yucca brevifolia*), amongst others (Schweiger et al., 2020).

Here we focus on the long-lived, iconic Clanwilliam cedar (*Widdringtonia wallichii* Endl. ex Carrière synonym *Widdringtonia cedarbergensis* Marsh), from the Cederberg Wilderness Area (Fig. 1). This species has special conservation importance due to its cultural significance, with the region taking its name, being one of the few trees in a shrub dominated landscape, as well as playing an important role in environmental education. Despite being among the most well-researched wild plants in South Africa (Richardson, 1993), it continues to decrease in abundance (White et al., 2016).

Adult mortality in this species has been well-studied, with continued decreasing population trends initially triggered by over-harvesting for timber, which likely began in at least the late 1700s (Hubbard, 1937; Smith, 1955). Even though the species is currently protected (Manders, 1986), it continues to suffer declines, likely due to increasing temperatures and the current fire regime (Sugden and Meadows, 1990; White et al., 2016) and is presently listed as critically endangered (Farjon et al., 2013).

The design of effective conservation measures for *W. wallichii* is complicated by a lack of information regarding its reproductive ecology, in particular around its recruitment, with little known about its seed biology and the role fire may play in its dispersal and germination. The primary hypotheses proposed to explain its decline are 1) frequent fires causing high juvenile mortality, 2) infrequent but high intensity fires that kill adult trees, and 3) excessive seed predation by small mammals (Manders, 1986).

1.1. Wildfire

Widdringtonia wallichii is an obligate reseeder and some consider it poorly adapted to its fire-prone environment because it cannot resprout, it is not serotinous, and it has a relatively slow growth and maturation rate. Serotiny and resprouting are two common traits in the species' Cape fynbos habitat (le Maitre and Midgley, 1992). Dominant species that co-occur with *W. wallichii*, such as *Protea nitida*, are both serotinous and capable of resprouting. Both of these traits also occur in *W. wallichii*'s widespread congener, *W. nodiflora*. Serotiny provides a mechanism for seed protection from fires as seeds are stored in woody cones and only released post-fire; *W. wallichii* seeds, however, are released annually and have no known adaptation to protect them from fire and predation. Adult *W. wallichii* can survive a degree of scorching but often die in fire (Manders, 1987a). This has led to a focus on factors which determine, or mitigate, adult mortality and an almost total absence of information on the seed to seedling aspect of the plant's natural history.

Manders (1987a) created a demographic model of the species that led him to note, "A paradoxical situation is presented by the results of the model in that wildfires - the natural situation - are found to be unacceptable because they now cause excessive losses...". The paradox is that the species occurs within the naturally fire-prone Cape vegetation of fynbos and therefore it is unlikely that fire per se can be unacceptable. Proteaceae dominate the canopy of Cape fynbos, including in cedar habitats, and most of these taxa are also obligate reseeders and die in fires. Key to Proteaceae management is not the



Fig. 1. Clanwilliam cedars (*Widdringtonia wallichii*) growing in Cederberg Sandstone Fynbos amongst rocks, two years post-fire, in the Middelberg region of the Cederberg Wilderness Area. Photograph taken in November 2018 (Copyright: J.D.M. White).

avoidance of fire, or of high fire intensity, but the maintenance of the recruitment process (Kruger, 1982; Kraaij et al., 2013). We suggest this applies to the cedar.

1.2. Cedar recruitment

An alternative explanation for the lack of recruitment could be that *W. wallichii* falls into the category of sporadically recruiting plants, such as many other long-lived plants (Schweiger et al., 2020). These plants may rely on occasional recruitment linked to long-term fluctuations in a certain environmental signal (e.g. increased rainfall or longer fire return intervals). This suggests that recruitment of *W. wallichii* may be highly variable and the current poor recruitment (White et al., 2016) may not be a conservation concern. However, without long-term (decadal or likely longer) studies of population dynamics this is difficult to discern and in light of this conservation managers must make informed decisions based on available information on important demographic drivers and patterns (Schweiger et al., 2020).

Manders (1987a) noted that prescribed burns in spring had reduced adult mortality of larger trees but had resulted in negligible recruitment. In contrast, summer wildfires led to high mortality in all size classes but significant recruitment from seed (Manders, 1987a). Although fire has been recognised as an important factor in *W. wallichii* recruitment (Manders, 1986; Mustart et al., 1995), this was thought to be due the removal of self-allopathic leaf litter, which inhibited germination (Manders, 1987a) without consideration of other important drivers of germination, such as heat and smoke.

Between 1956 and 1972, State Forest management applied an approach of frequent (every 4–6 years) low intensity fires in an attempt to reduce fuel loads to specifically protect adult trees throughout the species' range (Manders, 1986; Mustart et al., 1995). With short fire return intervals favouring adult survival there were low recruitment rates. This led to two approaches to boost seedling numbers: 1) direct seed sowing and 2) seedling out-planting.

Direct seed sowing was first attempted at the turn of the 19th century, but the practice was abandoned due to low germination rates, which were attributed to excessive rodent granivory (Manders, 1986). Direct seed sowing was only experimentally tested much later and resulted in seed germination at a rate of 10–30% of seeds within small mammal exclosures (Manders and Botha, 1987, 1989). Nonetheless, rodent granivory was still considered a hindrance to re-establishment efforts due to high seed removal rates (Botha, 1989). Large-scale seedling out-planting events were employed immediately post-fire to boost seedling numbers with little success (Mustart et al., 1995), although the long-term ecological viability of these efforts have not been evaluated.

1.3. Seed dispersal

Widdringtonia wallichii produces the largest seeds of all its congeners, and its seeds are exceptionally large among all southern hemisphere conifers (Ladd et al., 2013). The seeds were thought to be dispersed by wind and rain surface runoff (Andrag, 1977; Manders, 1987b), although this is unlikely as they lack wind-dispersal appendages, such as plumes or wings, and due the large size of the seeds they would be readily consumed or incinerated by fire without effective dispersal into the soil bank (Moore and Vander Wall, 2015; White et al., 2017).

Here we address two key knowledge gaps in the demographic mechanisms of *W. wallichii*'s seed dispersal and germination to help advance a more holistic conservation framework for the species. We ask: i) what is the seed dispersal pathway of seeds and ii) what stimulates germination? We hypothesized seeds of the species are scatterhoarded, similarly to most wingless non-serotinous large nuts in the Cape, such as *Willdenowia incurvata* and *Leucadendron sessile* (Weighill et al., 2017; White et al., 2017). We therefore predicted that germination is fire stimulated (either by heat and/or smoke; see Keeley and Bond, 1997). Higher germination was expected in burned stands where smoke and heat may provide cues for seed germination. Finally, we investigated the recent fire regime within the context of seed fall phenology to understand how the timing of fire and seed production may affect seed dispersal and germination.

2. Methods

2.1. Study area

We conducted our field research in stands of *W. wallichii* in the Cederberg Wilderness Area (CWA) of southwestern South Africa (−32.439277°, 19.188092°), an area dominated by fire-prone sclerophyllous heathlands known as fynbos. There are several co-occurring obligate reseedling species across the chosen field sites, such as *Leucadendron glaberrimum* and *Paranomous bracteolatus*, though the vegetation of the area is dominated by wind-dispersed, coppicing species, such as the overstorey species, *Protea nitida*, and understory, small-seeded members of the Restionaceae. *Widdringtonia wallichii* is endemic to the Cederberg Wilderness Area, which cover an area of approximately 710 km² of the Western Cape province of South Africa. It is restricted to a narrow elevational band, between 900 and 1600 m above sea level, on quartzitic, nutrient poor sandstone soils. Trees are generally restricted to cliffs, rocky outcrops and very rocky slopes (Manders, 1986). The seeds are wingless nuts, which are released between February and April (mass = 92.7 ± 8.27 mg [mean ± SE]; White et al., 2017).

Based on all fires in the CWA between 1956 and 1986, Brown et al. (1991) estimated a fire return interval of 11–15 years for the area. Later work incorporating data on post-fire ecosystem recovery trajectories suggested an estimated fire return interval of 18–21 years for the region (Merow et al., 2014), which may reasonably estimate historical fire regimes, as it

incorporates dominant vegetation characteristics in response to local climatic drivers. However, the fire regime of the CWA has been highly modified for several centuries and experimentally managed for the second half of the 20th century (Manders, 1986).

Widdringtonia wallichii seeds were collected from *ex-situ* sites in domesticated stands at De Rif plantation and Driehoek farm (Fig. 2) for use in seed trials. Seeds from Driehoek farm were used for germination experiments; only one seed source was used to standardize seed origin.

2.2. Seed dispersal

To understand the seed dispersal pathway, we placed groups of ten seeds at seed depots throughout the range of *W. wallichii* trees and observed interactions with potential dispersers and consumers using motion-sensitive camera traps (Ltl Acorn 6210M, Bushnell Natureview HD and Bushnell Trophy HD). These seed fate observations were recorded on seven visits at four sites over three separate years (2014, see White et al. (2017), 2018 and 2019) (Fig. 2). Three of the sites were used twice (Supplementary Table 1). Sampling effort varied between visits but included at least nine independent seed depots during each visit (see Supplementary Table 1).

To determine seed fate, we used seed depots placed close to adult trees. Depots consisted of ten to fifty seeds labelled with either 30 cm lengths of fluorescent thread or reverse-wound cotton bobbins attached to each seed using a quick setting glue. For the bobbins, thread was unwound from the spool as seeds were moved away from the depots, allowing the thread to be

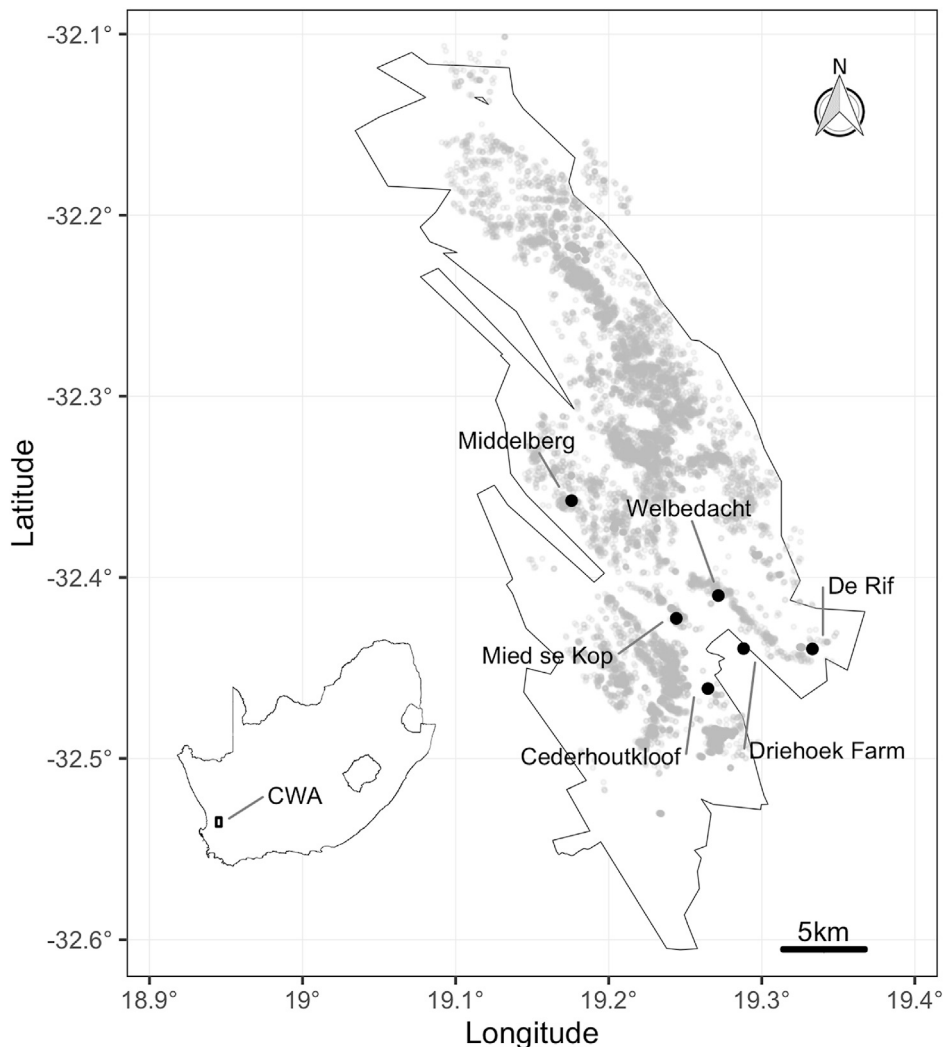


Fig. 2. Sites of seed collection, seed trials and seedling monitoring within the Cedarberg Wilderness Area (CWA). Gray points show the global distribution and locations of all large, adult *Widdringtonia wallichii* trees as recorded by Slingsby and Slingsby (2019a). Middelberg, Mied se Kop, Cederhoutkloof and Welbedacht were used as seed trial sites, Driehoek Farm and De Rif were used for seed collection, and seedling monitoring was done at De Rif.

tracked and seeds located. For fluorescent threads, we used ultraviolet-emitting light-emitting diode torches to locate seeds removed from seed depots at night. Searches for labelled seeds consisted of walking in a spiral outward away from seed depots for up to 10 m. Searches were expanded up to 15 m if no seeds were found within 10 m. Search groups consisted of at least three people with torches. Seed depots were set up in the late afternoon and were checked either before sunrise the next morning or directly after sunset the following day, allowing us to record a single night's seed interactions. Seed depots were checked daily and left for up to three nights. Once all seeds were removed from a depot, the depot was removed from the study. Distances of removed seeds and depths of buried seeds were recorded, and seeds were classified as consumed, buried or unrecovered (instances in which tags failed). Seed depots were placed at least 20 m from one another to reduce the possibility of pseudo-replication (i.e. one small mammal visiting several depots). Overall, we monitored 3090 seeds at 95 seed depots over the duration of the study. Comparisons in seed fate between interacting small mammals was not analysed statistically, due to small sample sizes.

Cameras were focused on seed depots from at least 1 m away and were set to record 30-second videos when triggered, with no time delay between recordings. An event was defined as an animal interacting with a seed (removed, consumed, explored, ignored or unknown). If the animal changed its behaviour, e.g., dropping a seed after inspecting and consuming a different seed, or entered/left the field of view, it was considered a new event. Additionally, if an animal entered the field of view and a seed was removed by the next recording, it was considered a removal event. Once all seeds were removed or consumed from a seed depot, no further events were documented. Small mammal identifications were based primarily on knowledge of species recorded in previous small mammal trapping (e.g. Botha, 1989), and body:tail ratio and overall body size, since most species in the region are readily identified using these criteria (Skinner and Chimimba, 2005).

2.3. Seed germination

An approximately 100 year old *W. wallichii* plantation at De Rif (see Fig. 2), with a relatively high density of cedar trees compared to wild populations, had a large portion of the plantation burned, and most adults were killed in a wildfire in December 2016. The fire burned approximately 24,138 ha of fynbos, largely within the CWA. This fire provided the opportunity to compare seed germination responses to wildfire in burned and unburned dense stands of trees. In December 2017, we counted all seedlings (defined as < 10 cm tall) in line transects through the burned (total area sampled; 265 m²) and non-burned (150 m²) portions of the plantation, noting adult presence and the number of seedlings. Additionally, to determine the depth from which seeds germinated, we extracted single seedlings occurring in groups of three or more (as only one of each group would likely survive) and measured the distance between the radicle end and the emergence level (soil surface; see Brits, 1987) and compared these with the depth of buried seeds around seed depots using a Kruskal-Wallis rank sum test.

To further test seed germination triggers, we conducted an *ex-situ* experiment in a glasshouse in April 2017. Treatments for the glasshouse experiment included planting (i) unmanipulated control seeds, (ii) scarified seeds, (iii) heated, or (iv) smoke-treated seeds. Before imbibing, we scarified or heated seeds. Scarified seeds had a small portion of the seed coat hand-sanded until exposing the endosperm, while heated seeds were placed 1 cm deep in a large tray of river sand in an oven at 80 °C or 100 °C for 5 or 10 min and then removed and left in the sand until completely cooled down. Time and temperature were crossed to create four levels of the heat treatment. Simulated fire temperatures and exposure times were selected based on values measured by Newton et al. (2006). Smoked seeds were imbibed in a smoke primer solution (Super Smoke Plus, South Africa; 20 seeds/smoke primer filter paper/50 ml water) for 24 h. All other seeds were imbibed in water for 24 h. All seeds were placed 1 cm deep in two trays per treatment type (20 seeds per tray) with 10 cm of river sand. Each tray was watered with 500 mL of water once weekly for the duration of the experiment (11 weeks). The experiment was run from May–July 2018, and trays were left in ambient conditions in the glasshouse with approximately a 10:14 h light:dark cycle. Temperatures ranged between 10 and 30 °C. We considered the presence of cotyledons breaking the soil surface as successful germination. We used 40 seeds in each treatment level. Germination was monitored twice weekly. Germination success between groups was compared with survival analysis and pairwise comparisons using log-rank tests with the Benjamini and Hochberg *P*-value adjustments in the 'survival' (Therneau, 2015) and 'survminer' R packages (Kassambara et al., 2019).

To test whether seed germination depth could be accurately estimated, we pre-treated seeds using the same procedure as heated seeds (10 min at 100 °C) and buried them at depths of 10, 20 and 40 mm in a separate tray for each treatment (*n* = 10 per treatment level). Seedlings were grown in a glasshouse. Once true leaves emerged, we extracted the seedlings and measured the distance between the radicle end and the emergence level (soil surface) of the seedling hypocotyl. Hypocotyl distance was then compared with the original burial depth using paired, two-sided *t*-tests.

2.4. Wildfire

To understand the context of how seed fall phenology could interact with fire, we used the spatio-temporal fire history, recorded by the regional conservation management agency, CapeNature, and provided by the national conservation management agency (South African National Biodiversity Institute, bgis.sanbi.org). The fire history spans 1944 to 2018 and we used these data together with a recently compiled collection of spatially accurate locations of *W. wallichii* tree locations identified with remote sensing techniques (13,419 trees; see Slingsby and Slingsby, 2019a and Slingsby and Slingsby, 2019b for the dataset). As these trees are all large and healthy (Slingsby and Slingsby, 2019a) and as the species' growth rate is slow (Hubbard, 1937), we make the assumption here that trees with canopy diameters greater than 4 m are at least 80 years old,

which spans the duration of the fire history dataset. This suggests that all fires over this time period would be ecologically relevant to the seed fall phenology of all trees in the dataset. We used these two datasets to determine the proportion of fires that *W. wallichii* adult trees experience by extracting fire information based on tree point locations and the intersecting fires. We 1) summed the overall number of fires for each month over the fire history period; 2) summed the overall number of fires for each month that was experienced by cedar trees; 3) determined whether fires occurred either before or after cedar trees' peak seed fall period in March. The fires before peak seed fall were considered unfavourable and described as early season (June to February), and those after were considered favourable and described as late season (March to May). The season of fire (early or late) was then summed for each *W. wallichii* tree and expressed as a percentage of early season fires relative to total fires experienced. 4) We then identified sites where the most recent fire experienced by cedar trees was an early season fire. 5) We used survival analysis to estimate the mean fire interval for each tree using survival regression models with a Weibull distribution (Johnson and Gutsell, 1994; Therneau, 2015). This method accounts for right-censored data, where we do not have a known interval value between the most recent fire event and the end of the recorded fire period (see Cyr et al., 2016). This method resulted in convergence for 12, 857 trees. We removed trees from our analysis that either did not converge or trees that had experienced only two fires during the recorded fire period, resulting in 12, 852 trees in the fire return interval analysis.

All data were processed, analysed and visualized using R programming language (R Core Team, 2019). Values are shown as mean $\pm$ standard error (SE), unless stated otherwise.

3. Results

3. 1 Seed dispersal

Seed removal was effectively bimodal, with 40% of depots ($n = 38$) having all seeds remaining, while 47% of depots ($n = 45$) had all seeds removed or consumed *in situ*. The overall mean proportion of seeds per depot removed from depots was 0.55 ± 0.05 , whereas the seed removal per depot where at least one seed was removed was 0.92 ± 0.03 . Overall, 60% of seeds were removed from depots, of which we located 61% during active searching. Almost all located seeds were consumed (96%, 1091 seeds), and the seed husks were found in large clumps, typically within 5 m of the seed depots. Only a small fraction of located seeds were buried (4%, 43 seeds), and those were found in caches 15 ± 1.55 mm deep.

Camera traps recorded 619 animal-seed interactions over 18 days of recording during seven visits to the CWA (see Supplementary Table 2 for summary of recorded interactions). Nocturnal small mammals were the only animals recorded that interacted with *W. wallichii* seeds. Three small mammal species removed seeds: *Micaelamys namaquensis* (Namaqua rock rat), *Acomys subspinosus* (Cape spiny mouse) and *Elephantulus edwardii* (Cape rock sengi). *Micaelamys namaquensis* dominated camera events (542 events, 88% of all events). They most often removed seeds from depots (300 events, 55% of total *M. namaquensis* events) and were regularly observed to consume seeds *in situ* (140 events, 26%) (See Supplementary Video 1). Cameras showed that *M. namaquensis* was the seed removal agent at 46 of 95 seed depots over the study; 66% of seeds (1066 of 1620) were consumed at those depots, and the remaining seeds were unrecovered 31% (497 of 1620). Only 3% of seeds (57 of 1620) were left intact at seed depots where *M. namaquensis* were active. At the seed depots where *M. namaquensis* was

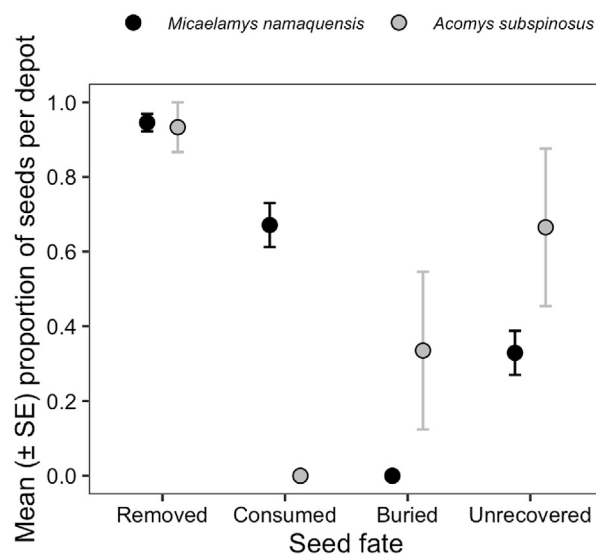


Fig. 3. The mean ($\pm$ SE) proportion of seeds per depot across seed fates (removed, consumed, buried or unrecovered) and interacting small mammal species for *Widdringtonia wallichii* dispersal.

observed the relative proportions of seed fates were removed 0.95 ± 0.02 , consumed 0.67 ± 0.06 , buried none and unrecovered 0.33 ± 0.06 (Fig. 3).

Supplementary video related to this article can be found at <https://doi.org/10.1016/j.gecco.2020.e01062>.

Acomys subspinosus was recorded in 47 events (8% of all events); 96% ($n = 45$) of those were seed removal events (See Supplementary Video 1). *Acomys subspinosus* was the seed removal agent at just three (of 99) seed depots. They were observed at Cederhoutkloof in May 2018 and at Welbedacht in November 2018. Twenty-nine % (43 of 150) of seeds were buried at those depots, while 65% of seeds (97 of 150) were unrecovered and 7% (10 of 150) were left intact at depots. No seed predation was observed. At the seed depots where *A. subspinosus* was observed the relative proportions of seed fates were removed 0.93 ± 0.07 , consumed none, buried 0.34 ± 0.21 and unrecovered 0.67 ± 0.21 (Fig. 3).

Elephantulus edwardii was occasionally observed to remove (three events) and explore (10 events) seeds. *Elephantulus edwardii* is predominantly insectivorous, but it is known to eat seeds in captivity (Skinner and Chimimba, 2005). Although previously recorded in the CWA (Botha, 1989), we found no evidence that the diurnal small mammal granivore, *Rhodomys pumilio*, was interacting with *W. wallichii* seeds.

3.2. Seed germination

3.2.1. Wildfire

In the early summer following the December 2016 wildfire, we recorded 241 seedlings in the old plantation. In the burned transect, we found 211 seedlings or $0.80/\text{m}^2$ versus 30 seedlings or $0.20/\text{m}^2$ in the unburned transect (χ^2 test, $\chi^2 = 53.16$, $\text{df} = 1$, $p < 0.001$). In the unburned area, the number of seedlings (plants < 100 mm tall) represents many years of establishment whereas in the burned area only seedlings that germinated after the 2016 fire were present. This effectively shows that post-fire recruitment is far greater than the sum of many years of inter-fire recruitment.

3.2.2. Glasshouse

Seeds heated for 10 min at 100°C began germinating after 24 days and all seeds germinated after 34 days, making this the most rapid germination treatment. Seeds heated for 10 min at 80°C first germinated after 25 days and all seeds had germinated after 41 days (Fig. 4). Seeds heated for 5 min at 80°C also began germinating after 25 days and after 43 days no more seeds germinated. Those seeds heated for 5 min at 100°C began germinating after 35 days and germination ended after 48 days (Fig. 4). Seeds in the smoke treatment also began germinating after 35 days, and all seeds germinated after 49 days (Fig. 4). The first scarified seeds germinated after 40 days and 75% had germinated by the time the experiment was ended. Seeds only treated with water began germinating after 43 days and 90% had germinated by the end of the experiment (Fig. 4).

The germination times were not significantly different between seeds heated for 10 min at either 80°C or 100°C , with both of these treatments showing the shortest time to germination and significantly different from all other treatments (Table 1). The next shortest germination times were found in the 5 min at 80°C treatment, which were significantly different from all other treatments (Table 1). Seeds exposed to smoke and seeds with heat for 5 min at 100°C were not significantly different from one another and had longer germination times than the aforementioned treatments, though were both significantly

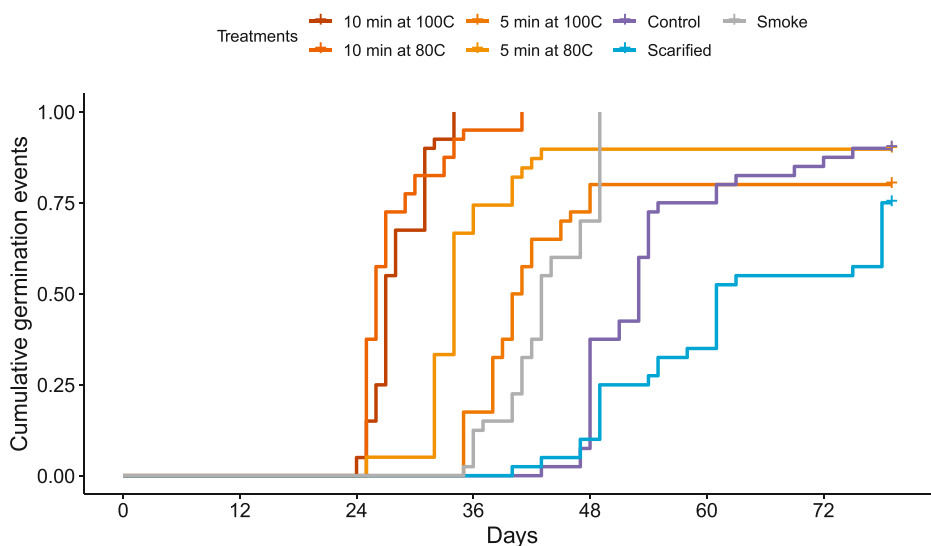


Fig. 4. Cumulative proportion of seeds germinated by treatment over the duration of the experiment.

Table 1Pairwise differences between seed germination rates by treatments using log-rank test. Values are *P*-values of pairwise differences.

	10 min at 100 °C	10 min at 80 °C	5 min at 100 °C	5 min at 80 °C	Control	Scarified
10 min at 80 °C	0.48					
5 min at 100 °C	<0.001	<0.001				
5 min at 80 °C	<0.001	<0.001	<0.001			
Control	<0.001	<0.001	0.0157	<0.001		
Scarified	<0.001	<0.001	<0.001	<0.001	0.0017	
Smoke	<0.001	<0.001	0.7603	<0.001	<0.001	<0.001

different and had shorter germination times than control and scarified seeds (Table 1). Control and scarified seeds were significantly different from one another and showed the second last and slowest germination times, respectively (Table 1).

Seedlings that emerged in the plantation germinated from seeds that originated from a soil seed bank and were buried at 19.24 ± 0.88 mm ($n = 21$). Seeds found buried by *A. subspinosus* were 15.00 ± 1.55 mm ($n = 35$) deep, which is significantly shallower (Kruskal-Wallis rank sum test, $\chi^2 = 4.87$, $df = 1$, $p = 0.027$).

When testing whether hypocotyl lengths can be used to estimate germination depth, we found that the depth at which we buried seeds matched for seeds buried at 10 mm (hypocotyl length = 10.38 ± 1.18 mm; 8/10 seedlings emerged; $t = -0.318$, $df = 8$, $p\text{-val} = 0.760$), but not for those at 20 mm (hypocotyl length = 24.80 ± 1.11 mm; 10/10 seedlings emerged; $t = -4.311$, $df = 9$, $p\text{-val} < 0.01$) and at 40 mm (hypocotyl length = 31.20 ± 2.20 mm; 5/10 seedlings emerged; $t = 4$, $df = 4$, $p\text{-val} < 0.05$). This suggests that hypocotyl length accurately represents germination depth at shallow depths but may become spurious at depths much greater than 20 mm.

3.3. Wildfire

Over the full fire history between 1944 and 2018, 119 fires intersected with *W. wallichii* trees. Of these 119 fires, 76% of fires occurred in the early season and 24% in the late season (Fig. 5a). When considering all of the trees that were intersected by the 119 fires, there were 66506 tree-fire events (Fig. 5b); 84% of fires were early season and 16% were late season. There was no evidence for a change in fire season over the recorded fire period (Supplementary Figs. 1 and 2).

Most *W. wallichii* trees (85%) experienced 70% or more of their fires in the early season, and 15% of trees experienced less than 70% of their fires in the late season (Fig. 6a). Eighty-one % of trees most recently experienced fire was early season, and 19% were most recently exposed to a late season fire (Fig. 6b).

The estimated overall mean fire return interval using the Weibull distribution for *W. wallichii* was 13 years for all trees; 74% of trees had a mean fire return interval of less than 14 years (Fig. 6c). As the number of fires increase, so the mean fire return interval decreases (Supplementary Fig. 3).

4. Discussion

4.1. Natural history

Widdringtonia wallichii was previously considered to be poorly fire-adapted and therefore prone to fire-induced extinction (Lückhoff, 1971). However, by studying the natural history of the plant, such as its seed dispersal and germination triggers, we come to a different conclusion.

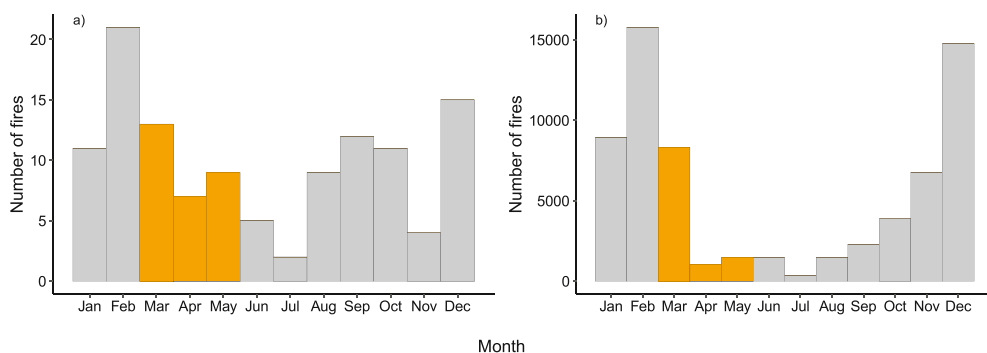


Fig. 5. Histogram of a) the frequency of fires in each month (1944–2018) in the Cederberg Wilderness Area that intersected with at least one *Widdringtonia wallichii* tree and b) all of the *W. wallichii* trees that intersected fires in each month over the full fire history 1944 to 2018. Orange bars show fires that occurred in a favourable period for effective seed recruitment (late season fires), while gray bars show fires that occurred outside of these favourable periods (early season fires). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

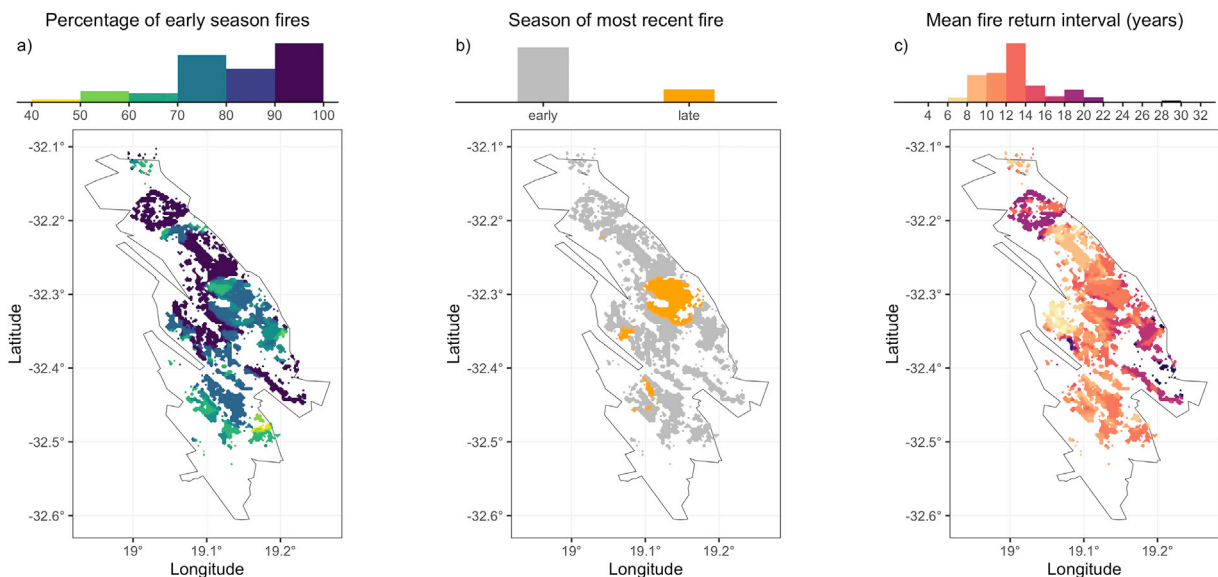


Fig. 6. Spatial representation of different fire-related metrics experienced by *Widdringtonia wallichii* trees, including a) the percentage of early season fires experienced by trees over the full fire history, b) the season of the most recent fire for each tree, and c) the Weibull estimated mean fire return interval experienced for each tree over the full fire history.

4.1.1. Seed dispersal

Two lines of evidence indicate that *W. wallichii* relies on scatterhoarding by *A. subspinosus*, which is a well-known scatterhoarder in the region (Midgley et al., 2002; Midgley and Anderson, 2005; Rusch et al., 2014; White et al., 2017), for effective seed dispersal: 1) camera and burial observations at seed trials and 2) *A. subspinosus* buries seeds at a depth from which seedlings can successfully emerge. Scatterhoarding is therefore a likely dispersal mechanism for *W. wallichii* seeds; it also explains how such large seeds enter the soil seedbank and avoid fire damage. However, seed burial only occurred in a few of our study sites; most seeds were lost to granivory by *M. namaquensis*, which is a known seed predator and does not hoard seeds (White et al., 2017). It is likely that *A. subspinosus* also consumed large proportions of seeds removed from seed depots, which may account for some of those seeds that were unrecovered. We speculate that burial only occurs where tree density is high enough to produce enough seeds to attract granivores and to satiate predation (such as at the De Rif plantation and Cederhoutkloof). The loss of large, old trees due to overharvesting may have set in motion the unexplained continued disruption and decline of the dispersal mutualism.

4.1.2. Seed germination

In the field and in the laboratory germination rate was enhanced by fire or fire derivatives (heat and smoke). In the cedars' case, seeds that germinate more rapidly after receiving the appropriate cues (fire-related and then responding rapidly to winter rainfall), would likely have a greater probability of survival, due to the short wet winter and prolonged dry summer. Seeds of hard-seeded species may benefit from heat-shock, which can trigger imbibition (Keeley and Bond, 1997). Both scatterhoarding (Midgley et al., 2002; White et al., 2017) and heat or smoke stimulated germination (Keeley and Bond, 1997) are common fire adaptations in the Cape, for protection from incineration and predation as well as a cue for the timing of germination for optimal recruitment.

Many previous *W. wallichii* researchers have recognised that germination is highest in the first season directly after late summer/autumn fires (Manders, 1986; Mustart et al., 1995), though the reasons and implications have not been fully understood. Manders (1987b) noted low germination and recruitment below adult cedar trees and wondered whether this was due to self-allopathy, as he recorded increased germination after cedar leaf litter was experimentally burned off. Manders (1987b) concluded that *W. wallichii* litter under mature canopies inhibits germination, and that the inhibiting factor was removed by fire. Given our results, the implication is that high germination after fire was due to heat-shock and/or smoke residues as a consequence of burning off the litter (a common Mediterranean ecosystem germination strategy) and not the removal of the self-allopathic litter itself.

Many curious aspects of *W. wallichii* biology can now be explained within its fire-adapted life history derived from the natural history of its dispersal. For example, Manders and Botha (1987) obtained low establishment (13%) in the field when seeds were added to non-burned sites, whereas we have achieved high germination after heat and smoke treatment. Botha (1989) later used small mammal exclosures and achieved better establishment rates (30%). However, their interpretation was linked to the paradigm of the time, that all seed removal by small mammals could be equated to seed predation. This paradigm discounted the potential dispersal benefits small mammals may have conferred to large-seeded plants. This

misunderstanding of the dual role of scatterhoarding small mammals as conditional mutualists as well as potential seed predators (Gómez et al., 2019) has pervaded seed research and maligned small mammals globally (Vander Wall, 1990). Such thinking may have overinflated the perceived impact of seed predation and discounted the crucial role small mammals, such as *A. subspinosus*, play in maintaining populations of large-seeded plants by dispersing seeds to safe sites, away from granivory and fire damage (Moore and Vander Wall, 2015). Similarly, the long-lived Joshua trees (*Yucca brevifolia*) of the Mojave Desert, rely on scatter-hoarding small mammals for effective directed dispersal into the soil bank, which reduces the probability of desiccation or granivory (Waitman et al., 2012).

4.1.3. Wildfire

The fire season experienced by *W. wallichii* trees since 1980 was concentrated in late spring and early summer (October–December). Seed release by *W. wallichii* is between late February and April, creating a mismatch between fire cues and seed germination. As seedbanks will be progressively depleted after seed-fall by granivorous small mammals, the greatest contribution to seedlings will come from the seeds dropped and dispersed in the few months preceding late summer/autumn fires. Small mammal populations greatly decline immediately postfire in fynbos (Midgley and Clayton, 1990; van Hensbergen et al., 1992). Thus, if fires continue to precede the natural seed fall period (i.e. early summer not late autumn) this will strongly negatively affect recruitment. Firstly, the latest years seed crop will be lost because it will not be mature by early summer and these seeds will be lost from trees that die in the fire. Secondly, for mature trees which survive summer fires, their seeds will either not be dispersed due to post-fire small mammal population declines. Also as these seeds are released in autumn, some months after fires have taken place in summer these seeds will not receive the required fire-related cue for germination. Both the lack of germination cues and lack of scatterhoarding will depress recruitment.

The preferred fire season should therefore not be late spring (Nov) to mid-autumn April (Brown et al., 1991) but rather after seed release and before winter rains i.e. from April/May/June, to better match the cedars seed fall phenology. This allows for the most recent years seed crop, the largest since the previous fire, to be effectively dispersed (scatterhoarded) and then receive fire-related germination cues before the winter rains.

Additionally, the time to first cones (14 years) and to full reproductive maturity (40+ years) greatly exceeds the current fire return intervals that most *W. wallichii* trees experience, creating an immaturity risk (Keeley and Fotheringham, 2000), where seedlings and juvenile trees will not successfully reach their adult stage. *Widdringtonia wallichii* are somewhat buffered by local fire extirpation, by having long-living adults in rocky, fire refugia (White et al., 2016), though recruitment is strongly suppressed and large, intense fires can also kill long-lived adults (Manders, 1986).

Since serotiny is ancient in conifers (He et al., 2016) and *Banksia* (He et al., 2011), this suggests *W. wallichii* is a more recent species that has lost the serotiny that is seen in its Cape congener *W. nodiflora*, and *W. wallichii* rather has evolved scatterhoarding. This trend from serotiny to scatterhoarding is also seen in another Cape lineage, *Leucadendron*, where scatterhoarding is derived (Tonnabel et al., 2014). A recent molecular phylogeny of the gymnosperms supports the view that *W. wallichii* is the result of a very recent speciation event (0.19 Myr, 100% bootstrap value) and that its clade stems from a more dated event (1.38 Myr, 100% bootstrap value) from the ubiquitous resprouting, small seeded and serotinous Cape species *W. nodiflora* lineage (Forest et al., 2018). This is further support that the reseedling, scatterhoarded and large seeded *W. wallichii* is likely to be highly adapted to its environment; both in terms of fire and climate, as in other fynbos species. *Widdringtonia wallichii* is a long-lived fire survivor and we interpret its decline as being due to a lack of sufficient recruitment, due to weak dispersal rates, mismatches in seed fall phenology and wildfires, and short fire return intervals killing juveniles, rather than excessive rates of adult mortality.

4.2. Conservation implications

Our findings on the proportion of seeds that are consumed vs. cached and the relative frequency of interactions between *W. wallichii* seeds and antagonistic vs. mutualistic small mammals is a major concern for effective seed dispersal and sustainable *W. wallichii* populations. Particularly given the high rates of seed dispersal of other nut-fruited plants by *A. subspinosus* in the region (Midgley et al., 2002; Rusch et al., 2014). To promote *W. wallichii* recruitment, the interaction between the mutualistic *A. subspinosus* and *W. wallichii* seeds needs to be further promoted. It is possible this small mammal has experienced a population decline in this region, as it is thought to be sensitive to short fire return intervals, likely due to its seasonal reliance on the large seeds of obligate reseedling plants (Palmer et al., 2016). Botha (1989) found a greater density of *A. subspinosus* than *M. namaquensis* using live-trapping. We found *A. subspinosus* rarely visited seed depots. The granivore *R. pumilio*, was surprisingly rare on camera traps, whereas Botha (1989) trapped several live individuals, though in low densities. While we did not conduct formal small mammal population assessments, this may indicate that the small mammal community of the region may have changed. Our previous work indicates the *M. namaquensis* is a generalist; for instance it is a frugivore (White and Midgley, 2017). It is possible that the decline in the densities and removal of larger individuals of nut-producing *W. wallichii* has led to decline in small mammal granivores and disrupted seed dispersal processes (Cordeiro and Howe, 2003).

It's important to not assess the life history of *W. wallichii* in isolation, but recognize it as a member of a fire-prone plant community. As with many conservation efforts, charismatic species draw far more public attention and consequently funding than other species. This may often serve to benefit the charismatic species and an umbrella effect may serve to benefit other

co-habiting species if management interventions are broadly applied. However, when management interventions prioritise some focal species over others, it may inadvertently be to the detriment of the focal species and the broader plant community.

Short fire return intervals, as practised between the 1950s and 1970s, with a rotation of cool burns every 4–6 years to promote adult tree longevity (Manders, 1986), would have decimated populations of obligate reseeders (and small mammals) that are part of the *W. wallichii* plant community. *Widdringtonia wallichii* is one of several large-seeded obligate reseeders found in the CWA, which are all likely to be scatterhoarded by small mammals. However, unlike the other reseeders, scatterhoarded plants *W. wallichii* is the only species with long-lived adults that can survive fire in protected refugia. All other species outside of these refugia will experience stand-clearing mortality and without their immaturity risk properly accounted for, will become locally extirpated (Keeley and Fotheringham, 2000), potentially leading to a landscape trap, where the landscape may be shifted and maintained in an altered disturbance regime (Lindenmayer et al., 2011). This community includes several Proteaceae species with large nut-like fruits, such as *Leucadendron glaberrimum*, *L. concavum*, *L. dubium*, *L. pubescens* and Restionaceae species such as *Willdenowia incurvata*, which are all likely obligate rodent-dispersed plants. Additionally, there are many large-seeded species, which may primarily rely on ants for seed dispersal, including *Leucadendron nervosum*, *Paranomous* spp, *Hypodiscus* spp, *Cannamois* spp but would provide a food source to small mammal granivores. A conservation approach that accounts for the general health of the plant community will likely benefit *W. wallichii* in the long run as it will increase the annual seed crop (of all plants), benefiting specialists seed dispersers such as *A. subspinosus* over generalist species.

The scatterhoarding perspective provides a mechanism for how over-harvesting alone could have led to further *W. wallichii* declines. There is no doubt that *W. wallichii* were heavily utilised and that the largest trees, which would disproportionately contribute to seed production, were the first to be extracted (Smith, 1955). Several large trees still have saw-mark scars. However, it is still not clear why the present populations cannot maintain themselves. The hypothesis that previously dense populations suppressed the understorey vegetation and caused fire suppression is unlikely (Manders, 1986); *W. wallichii* leaves are mere scales and the canopies are sparse. From our results, we infer that sparse populations of *W. wallichii* had lower degrees of small mammal satiation or even that sparse populations led to a change in small mammal composition from granivores (such as *A. subspinosus* and *R. pumilio*) to omnivores such as *M. namaquensis*.

Seed germination is greatly enhanced by fire and generally there is a delay between when seeds receive a fire cue, whether smoke or heat, and when the seeds germinate. The former is typically in summer-autumn and the latter in winter. This implies that seeds retain the stimulatory heat impact and therefore that seed sowing, as previously attempted by (Manders and Botha, 1987), should be attempted with heat treated seeds buried about 2 cm deep. This should especially take place near large rocks and in dense patches (typically where *A. subspinosus* buries them) and planted out from May to June. Large rocks provide some shelter from fire as well as better water relations (from shade and water shedding) (Mustart et al., 1995). This seed planting should also be restricted to the season immediately postfire, giving seedlings the best chance to grow to maturity before the next fire. This will be significantly cheaper and more biologically relevant than planting seedlings, since seedlings cannot practically be carried in significant numbers to the upper rocky slopes that is their typical environment. Seedling out-planting days likely have several benefits, including local community engagement, conservation education opportunities, increasing management visibility and potentially opening doors to additional conservation funding. Seeds and seedlings should be densely planted, and other endemic large-seeded species should be considered in this approach. This will provide the opportunity for seed-fall patches to become dense enough to recreate and sustain the mutualism.

We suggest that *W. wallichii* recruitment seems highly adapted to a fire regime with mostly early summer fires, as its dispersal mechanism provides a route to enter the soil bank, protecting seeds from the harmful effects of fire, while maintaining the cues required for germination. Currently, long-lived adults persist in rock outcrops (White et al., 2016), where fuel loads decrease, structural barriers increase and consequently, risk of fire-induced mortality is reduced. However, fire-related cues, such as smoke, would still likely be present as these sites still maintain some of their fuel load. Whether recruitment is sporadic, as suggested by Schweiger et al. (2020) remains unclear and will require long-term monitoring to infer.

This study primarily investigated the seed germination triggers and seed dispersal of *W. wallichii*. An assumption that remains to be conclusively examined is whether there is in fact a disrupted mutualism between *A. subspinosus* and *W. wallichii*. Though we were able to provide evidence that *W. wallichii* seeds are scatter-hoarded, seed fate experiments were conducted over a short-time period seeds. Follow-up studies should explore this interaction over longer time periods, as well as provide focused monitoring of *A. subspinosus* and other small mammal populations to allow direct comparisons with previous live-trapping studies (such as Botha, 1989).

In conclusion, this study provides an example of the benefit of understanding key demographic knowledge gaps, such as seed germination and dispersal, to understand recruitment limitations of a long-lived species. By better understanding the reproductive ecology of long-lived species, we can provide alternative approaches to managing wild populations by focusing on key aspects of their biology. This may allow species to buffer against future declines by providing recruitment opportunities.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.gecco.2020.e01062>.

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